

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	DSA0607	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization (BL5)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	30 (6 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).</i>		
Student Name:	V.SURYA SATVIKA	Reg. No.:	192524254
Section / Batch:	2025/D	Date:	

Code of Conduct

I _____ VANUMU SURYA SATVIKA(192524254)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ V.S.SATVIKA _____

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Scatterplots & Correlation Analysis	Correlation, outliers, and relationship identification using scatterplots	1	5
Correlograms & Feature Selection	Correlation matrices and feature importance analysis	2	5
Dimension Reduction (PCA, t-SNE, UMAP)	High-dimensional data visualization and comparison of techniques	3, 4	10

Bubble Charts & Multivariate Visualization	Encoding multiple variables and interpreting limitations	5	5
Hexbin Plots & Dense Data Visualization	Handling overplotting in large datasets	6	5
GRAND TOTAL:		30 Marks	

Annexure A – Short Answer Test

1. How does a violin plot provide more information than a box plot when analyzing data distribution?

A violin plot provides more information than a box plot because it shows both the summary statistics and the shape of the data distribution. A box plot mainly shows the median, quartiles, and outliers, whereas a violin plot also shows the density of the data.

a) What does the width of a violin plot represent?

The width of a violin plot represents the density of observations at a particular value. A wider section indicates a higher concentration of data, while a narrower section indicates fewer observations.

b) How can a violin plot help identify multimodal distributions?

A violin plot can identify multimodal distributions by showing multiple peaks or bulges in its shape. Each peak represents a concentration of observations around a particular value. Therefore, it helps identify whether the data has more than one mode.

2. How does a pair plot help in exploring relationships among multiple variables in a dataset?

A pair plot helps explore relationships among multiple variables by displaying scatterplots for every pair of numerical variables in a dataset. It provides an overall view of relationships, correlations, and distributions between variables.

a) What information is displayed along the diagonal of a pair plot?

The diagonal usually displays the distribution of each individual variable, commonly using a histogram or density plot.

b) How can pair plots assist in detecting clusters and correlations?

Pair plots help identify correlations by showing patterns between pairs of variables. An upward pattern indicates positive correlation, while a downward pattern indicates negative correlation. Groups of points appearing close together can indicate clusters.

Thus, pair plots are useful for identifying relationships, correlations, and clusters among multiple variables.

3. What is the purpose of a heatmap in data visualization and analysis?

A heatmap is a data visualization technique used to represent numerical values using different colors or color intensities. It helps in quickly identifying patterns, relationships, and unusual values in a dataset.

a) How are values represented in a heatmap?

Values are represented using different colors or shades. Different color intensities correspond to different numerical values. A color legend is generally provided to understand the values represented by each color.

b) Why are heatmaps useful for identifying patterns and anomalies?

Heatmaps make patterns easy to recognize because similar values are represented by similar colors. Unusual values or anomalies can be identified when a cell has a noticeably different color from surrounding cells. Heatmaps are therefore useful for quickly analyzing large amounts of data.

4. How does a parallel coordinates plot visualize high-dimensional data?

A parallel coordinates plot is used to visualize high-dimensional data on a two-dimensional surface. It represents several variables simultaneously, allowing different observations to be compared across multiple variables.

a) What does each vertical axis represent?

Each vertical axis represents a different variable or feature in the dataset. The axes are arranged parallel to each other.

b) How can this plot be used to compare observations across multiple variables?

Each observation is represented by a line that crosses all the vertical axes according to its value for each variable. By following a line across the axes, we can compare one observation across multiple variables. This helps identify similarities, differences, patterns, and groups in the data.

5. How does a contour plot represent three-dimensional data on a two-dimensional surface?

A contour plot represents three-dimensional data on a two-dimensional surface by using contour lines to show changes in the value of a third variable. The two axes represent two variables, while the contour lines represent the value of the third variable.

a) What do contour lines indicate?

Contour lines connect points having the same value of the third variable. Closely spaced contour lines indicate a rapid change in values, while widely spaced lines indicate a gradual change.

b) In which applications are contour plots commonly used?

Contour plots are commonly used in geographical and topographical mapping, weather and climate analysis, engineering, heat distribution analysis, and scientific applications.

6. How does a density plot help in understanding the distribution of continuous data?

A density plot helps in understanding the distribution of continuous data by displaying a smooth curve that represents the estimated distribution of the data. It helps to identify the concentration and spread of observations.

a) How is a density plot different from a histogram?

A histogram divides the data into bins and represents the frequency using bars. A density plot uses a smooth continuous curve to represent the estimated distribution of the data.

b) What does a peak in a density plot indicate about the data?

A peak in a density plot indicates a region where the data has a high concentration of observations. A higher peak means that values around that region occur more frequently.

Rubrics for Calculation

Weightage Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Criteria
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Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Conceptual Understanding	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding	Conceptual Understanding	30%
Accuracy of Answer	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer	Accuracy of Answer	25%
Application of Knowledge	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application	Application of Knowledge	20%
Completeness of Response	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response	Completeness of Response	15%
Clarity & Presentation	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand	Clarity & Presentation	